ENGINEERING GEOLOGY

UNIT - I : INTRODUCTION

Branches, Importance, Weathering, Geological Agents & Rivers

Syllabus (as given)

- Branches of Geology, Importance of Geology in Civil Engineering with case studies, weathering of rocks, Geological agents, weathering process of Rock, Rivers and geological work of rivers.

1. Introduction to Geology

Geology (Greek: geo = earth, logos = discourse) is the science that deals with the origin, composition, structure, history and processes of the Earth.

Engineering Geology = application of geological knowledge to civil engineering works (dams, tunnels, bridges, roads, buildings, reservoirs, canals, slopes).

- Earth facts: age ≈ 4.6 billion years; radius ≈ 6371 km; layers = **crust, mantle, core**
  - Crust: 5-10 km (oceanic), 30-70 km (continental) - mostly granite (SIAL) and basalt (SIMA)
  - Mantle: ≈ 2900 km thick; Core: outer (liquid) + inner (solid, Fe-Ni)
- Geology is an **observational + interpretative** science - rocks are studied in the field, in the lab and by instruments.

1.1 Branches of Geology

1. **Physical Geology** - study of the earth's materials, structure and the processes acting on and within it (weathering, erosion, volcanism, earthquakes).
2. **Mineralogy** - study of minerals: composition, crystal form, properties, origin, uses.
3. **Petrology** - study of rocks: their origin, texture, classification (igneous, sedimentary, metamorphic).
4. **Structural Geology** - deformation of rocks: folds, faults, joints, unconformities.
5. **Stratigraphy** - study of layered rocks, their sequence, age and correlation.
6. **Palaeontology** - study of fossils (ancient life) - used for dating and correlation of

strata.

7. **Historical Geology** - history of the earth and its life through geological time.
8. **Geomorphology** - origin and evolution of landforms (hills, valleys, plains, deltas).
9. **Economic Geology** - ores, coal, petroleum, building stones and other useful materials.
10. **Hydrogeology** - occurrence, movement and quality of ground water.
11. **Engineering Geology** - geology applied to engineering projects and site safety.
12. **Geophysics, Geochemistry, Seismology, Volcanology, Oceanography** - supporting branches.

Quick memory trick

- Minerals -> Rocks -> Structures -> Strata -> Fossils -> Landforms -> Water -> Engineering. (Material first, then arrangement, then time, then use!)

1.2 Importance of Geology in Civil Engineering

Every civil structure rests **on or in** the earth, so its safety depends on the nature of the ground. Geology helps the engineer in:

- **Site selection & investigation** - for dams, reservoirs, tunnels, bridges, highways, airports, power plants.
- **Foundation conditions** - bearing capacity, weak zones, cavities, soft clay, expansive (black cotton) soil, settlement.
- **Construction material** - identifying good sources of stone, sand, gravel, clay, lime, laterite for aggregate and bricks.
- **Water problems** - ground-water level, seepage, leakage from reservoirs, water supply and drainage.
- **Natural hazards** - earthquakes, landslides, floods, subsidence, tsunami: zoning and mitigation.
- **Structures in rocks** - attitude of beds, joints, faults decide stability of slopes, tunnels and dam abutments.
- **Cost & time control** - early geological knowledge avoids redesign, failures and extra expenditure.
- **Mineral & resource exploration** - roads/mines/quarries planning, geophysical surveys.

Remember

- Geologist finds out WHAT the ground is, the engineer decides HOW to build on it. "Geology is the foundation of foundations."

1.3 Case Studies (failures due to geological neglect)

(a) Vajont Dam, Italy (1963)

- 262 m high **double-curvature arch dam** - the dam itself survived, but a huge **landslide** (~270 million m^3) fell into the reservoir from the slope.
  - Cause: bedding dipping **towards the reservoir**, clay layers, old slide mass, rise in reservoir level and heavy rain reduced shear strength.
  - Result: giant wave overtopped the dam, ≈ 2000 lives lost. **Lesson:** study reservoir-rim slope stability.

(b) St. Francis Dam, California, USA (1928)

- 62 m **concrete gravity dam** failed within hours of first full filling; ≈ 430 people died.
  - Cause: the left abutment had old landslide mass and **weak schist**; the right side had **conglomerate that softened when wet** (slaking, uplift).
  - **Lesson:** rock type and its behaviour in contact with water must be tested.

(c) Malpasset Dam, France (1959)

- Thin **arch dam** on gneiss; abutment had a hidden **fault / foliation surfaces**. Water pressure (uplift) pushed a rock wedge out - ≈ 420 deaths.
  - **Lesson:** discontinuities (joints, faults) in abutments govern arch dam safety.

(d) Teton Dam, Idaho, USA (1976)

- Earth-fill dam failed on first filling: **seepage through fissured volcanic rock** (rhyolite) and the key trench caused piping and a breach.
  - **Lesson:** grouting and cut-off for jointed foundation rock; monitor seepage.

(e) Koyna Dam (Maharashtra, India) - 1967

- Reservoir-induced earthquake (M 6.5) near a **gravity dam** on Deccan basalt - dam was damaged but did not collapse.
  - **Lesson:** reservoir loading can trigger seismicity (RIS); seismic design is essential.

(f) Konkan Railway & Himalayan roads (India)

- Konkan Railway (Mumbai-Mangalore): many tunnels and cuttings in laterite, basalt and soft ground; **geological mapping** reduced slope failures and tunnel collapses.

- Himalayan highways: landslides on weathered, sheared and jointed rocks each monsoon - slope-cutting without geology = repeated failures.

Project	Geological issue	Lesson
Vajont (1963)	Bedding slope towards reservoir	Check rim stability
St. Francis (1928)	Weak schist, soft conglomerate	Test rock in water
Malpasset (1959)	Fault + foliation at abutment	Map discontinuities
Teton (1976)	Seepage in fissured rock	Grouting / cut-off
Koyna (1967)	Reservoir-induced quake	Seismic design

2. Weathering of Rocks

Weathering = the in-situ (in-place) breakdown of rocks at or near the earth's surface by the action of atmosphere, water and organisms **without transportation of the material.**

- Weathering is **different from erosion**: erosion = weathering + removal/transport by agents (water, wind, ice).
- Result: hard rock -> loose fragments -> **soil** (regolith).
- Rock weathering profile: fresh rock -> partially weathered rock -> saprolite -> residual soil.

2.1 Types of Weathering

1. **Physical (mechanical / disintegration)** - rock breaks into pieces; no change in chemical composition.
2. **Chemical (decomposition)** - minerals are altered and new minerals formed.
3. **Biological (organic)** - plants, animals, man and micro-organisms help both of the above.

2.2 Physical Weathering Processes

- **Temperature changes (insolation weathering)**: day heating - night cooling -> repeated expansion/contraction -> cracks. Common in **deserts**.
- **Exfoliation (onion-skin weathering)**: outer shells peel off like onion layers; seen in granite domes.
- **Frost action / ice wedging**: water enters cracks, freezes, volume increases ≈ 9% ->

prises rock apart. Cold regions.

- **Salt crystallisation**: salts crystallise in pores and grow - splitting rock (arid and coastal zones).
- **Pressure release (unloading)**: erosion of overlying rock reduces confining pressure -> sheet joints form.
- **Wetting and drying (slaking)**: shales, clays swell and shrink -> flaking.
- **Abrasion by wind, water, ice** - sometimes grouped here.

2.3 Chemical Weathering Processes

- **Oxidation**: oxygen reacts with Fe-minerals -> iron oxides (rust-brown limonite, red hematite). Example: $4FeO + O_2 \rightarrow 2Fe_2O_3$.
- **Hydration**: minerals absorb water and swell - anhydrite -> gypsum; hematite -> limonite.
- **Hydrolysis**: reaction of minerals with water (H+ and OH-); **feldspar -> clay (kaolinite)**; the most important process on silicate rocks.
  - $2KAlSi_3O_8 + 2H_2O + CO_2 \rightarrow Al_2Si_2O_5(OH)_4 + K_2CO_3 + 4SiO_2$
- **Carbonation**: CO_2 dissolved in rain makes carbonic acid H_2CO_3; it dissolves **limestone** -> caves, sinkholes. $CaCO_3 + H_2CO_3 \rightarrow Ca(HCO_3)_2$
- **Solution**: direct dissolving of soluble minerals (rock salt, gypsum, calcite).
- **Reduction**: removal of oxygen under waterlogged conditions ($Fe^{3+} \rightarrow Fe^{2+}$), grey-green colours.

2.4 Biological Weathering

- **Roots** of trees grow in joints and wedge rock apart (root wedging).
- **Burrowing animals** (worms, termites, rodents) mix and expose fresh surface.
- **Lichens, bacteria, fungi** secrete organic acids -> decompose minerals (chelation).
- **Man** - mining, quarrying, blasting, agriculture, construction speed weathering.

2.5 Factors Controlling Weathering

- **Climate**: temperature and rainfall - fastest in hot-humid regions (chemical); frost in cold regions (physical).
- **Rock type & mineral composition**: quartz resists; olivine, calcite, feldspar weather quickly.
- **Structure**: joints, bedding, faults give water entry paths and larger surface area.

- **Topography / slope**: steep slopes remove debris (thin soil); gentle slopes keep it (deep weathering).
- **Vegetation, time & ground water.**

Physical	Chemical	Biological
Mechanical break-up	Change in composition	Organisms
Dry / cold climate	Warm and wet climate	Any climate with life
Gives angular debris	Gives clay, solutions	Humus, soil

2.6 Importance of Weathering in Civil Engineering

- Weathered rock has **low strength and high permeability** - lowers bearing capacity; foundations must go below the weathered zone.
- Depth of weathering controls excavation depth, slope angle and cost of tunnelling.
- Weathered stones are **unsuitable** as building stone / aggregate (reduced durability).
- Weathering produces **soils** - agriculture, clay for bricks and earth dams; **laterite** and **bauxite** are weathering products.
- Causes landslides, rock falls, and decay of monuments and concrete.

2.7 Grades of Weathering (engineering classification)

Grade	Description
I - Fresh	No visible sign of weathering; rock sound and ringing sound on hammer blow
II - Slightly	Discolouration along joints; strength almost same as fresh rock
III - Moderately	Less than half of rock decomposed; weak, hammer gives dull sound
IV - Highly	More than half decomposed; can be broken by hand
V - Completely	All rock turned to soil but original texture visible
VI - Residual soil	Fully soil, original rock texture destroyed

- Foundations of heavy structures should rest on **Grade I-II rock**; Grade IV-VI need removal, grouting or deep piles.

2.8 Soil Formation (Pedogenesis)

- **Soil** = weathered rock + organic matter + air + water. Formed by weathering over long time under climate, parent rock, relief, organisms and time (**CLORPT**).
- **Soil profile (horizons):** O (humus) -> **A** (top soil, leached) -> **B** (sub-soil, accumulation) -> **C** (weathered parent rock) -> **R** (bed rock).

- Indian soils: **alluvial** (river plains), **black cotton / regur** (Deccan basalt; swells, shrinks, bad for foundations), **red** (granite, gneiss), **laterite** (hot, humid), **desert** and **mountain** soils.
  - Anantapur region (A.P.): mostly **red soils & black cotton soils** over granite-gneiss and Cuddapah rocks.
- **Residual soils** keep the structure of the parent rock; **transported soils** are loose and heterogeneous.

2.9 Relative Resistance of Minerals to Weathering

- Least resistant -> **olivine, pyroxene (augite), Ca-plagioclase, calcite** (form first at high temperature, unstable at the surface).
- Moderate -> hornblende, biotite, orthoclase.
- Most resistant -> **muscovite, quartz** (stable, forms sand, quartzite).
- Rule (Goldich series): minerals which crystallise first from magma weather first.

Common weathering products

- Feldspar -> clay (kaolinite) | Mica -> clay + iron oxides | Limestone -> solution, caves | Basalt -> black cotton soil / laterite | Granite -> red soil, sand

3. Geological Agents

Geological agents are natural forces that change the earth's surface and interior. They work continuously but slowly ("the present is the key to the past" - **Uniformitarianism**, James Hutton).

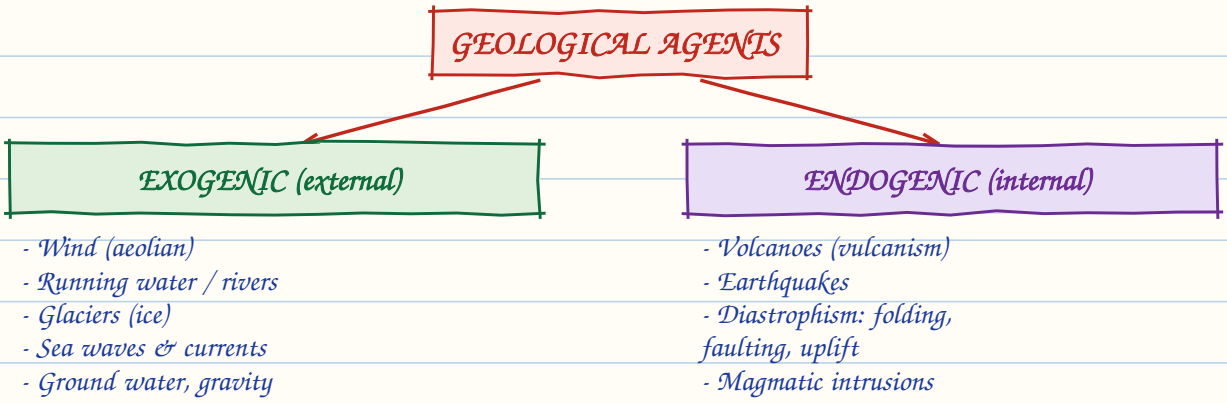


Fig 1: Geological agents

- **Gradation** = levelling of land by agents; has two parts: **degradation** (wearing down - weathering, erosion, transport) and **aggradation** (building up - deposition).
- Sequence of work of any external agent: **Weathering -> Erosion -> Transportation -> Deposition -> Compaction / Lithification.**

3.1 Work of the main agents (summary)

Agent	Erosional work	Typical landforms
River	Down-cutting, abrasion	V-valley, gorge, delta
Wind	Deflation, abrasion	Dunes, loess, mushroom rock
Glacier	Plucking, abrasion	U-valley, moraine, cirque
Sea waves	Hydraulic action	Cliffs, beaches, stacks
Ground water	Solution	Caves, sinkholes, stalactites

- **Engineering relevance**: agents decide soil type (alluvial, aeolian, glacial, marine), foundation problems, silting of reservoirs, coastal erosion and the shape of valleys used for dams.

4. Rivers and Geological Work of Rivers

A **river** is a natural stream of water flowing in a definite channel from higher to lower ground under gravity. **Rivers are the most powerful agent of denudation** on land, shaping valleys, plains and deltas.

- Parts of a river: **source (head)** -> **upper course** -> **middle course** -> **lower course** ->

mouth (sea / lake).

- **Tributaries** join the main stream; **drainage basin (catchment)** = area drained; **watershed / divide** separates two basins.
- Drainage patterns (seen on maps/aerial photos): **dendritic** (tree-like, uniform rock), **trellis** (parallel, folded strata), **radial** (volcano / dome), **rectangular** (joints/faults), **parallel** (steep uniform slope).

4.1 Erosion by Rivers

River erosion depends on **velocity, volume (discharge), slope, load carried and rock resistance.**

1. **Hydraulic action** - sheer force of moving water dislodges loose rock; compresses air in cracks.
2. **Abrasion (corrasion)** - rock fragments carried by water scrape the bed and banks like sandpaper; main erosional process; makes **potholes**.
3. **Attrition** - boulders collide and get **rounded and smaller** as they travel downstream.
4. **Solution (corrosion)** - soluble rock (limestone) dissolves in river water.

- Types of erosion by direction:
  - **Vertical (down-cutting)** - deepens valley in upper course -> V-shaped valley, gorge, canyon.
  - **Lateral (side-cutting)** - widens valley, forms meanders in middle/lower course.
  - **Headward erosion** - river extends upstream at its source; leads to **river capture (piracy)**.

Velocity facts

- Erosive power increases with the SIXTH power of velocity (v^6 law) - doubling velocity gives ≈ 64 times more load-carrying capacity! Floods therefore do most of the geological work.

4.2 Transportation by Rivers

- **Solution** - dissolved salts (Ca, Mg, Na, bicarbonates) carried invisibly.
- **Suspension** - clay, silt, fine sand held up by turbulence; makes river muddy.
- **Saltation** - sand grains hop along the bed in jumps.
- **Traction** - pebbles and boulders roll, slide, drag on the bed (**bed load**).
- **Competence** = largest particle a river can move; **Capacity** = total load a river can

carry.

4.3 Deposition by Rivers

River deposits load when **velocity falls**: entry into sea/lake, flattening of slope, reduction in discharge, widening of channel, or obstruction. Deposits are called **alluvium** (sand, silt, clay, gravel).

4.4 Landforms made by Rivers

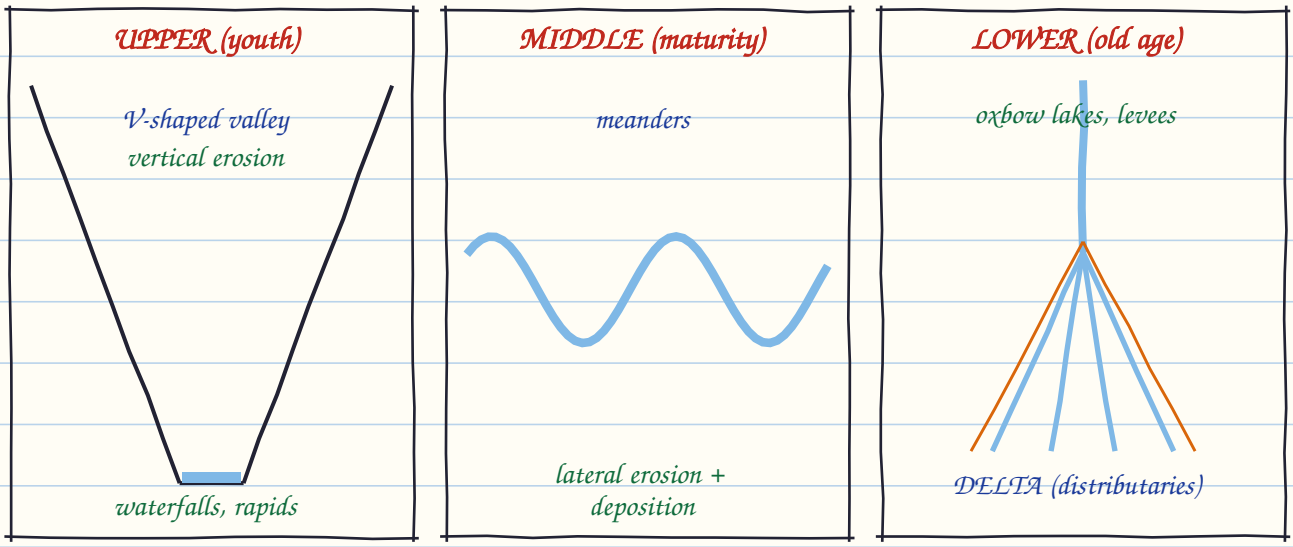


Fig 2: Three courses of a river and their landforms

Erosional landforms

- **V-shaped valley** - narrow, steep sides; typical of youthful stage.
- **Gorge & canyon** - deep narrow valley cut in hard rock (Grand Canyon; Gandikota gorge on River Penna, A.P.).
- **Waterfall & rapids** - hard rock band over soft rock; undercutting forms **plunge pool** (e.g., Jog Falls, Hogenakkal).
- **Potholes** - cylindrical holes drilled by swirling pebbles.
- **Interlocking spurs** - ridges alternate from the sides of a V-valley.

Depositional landforms

- **Alluvial fan / cone** - fan of debris at the foot of a mountain where slope suddenly decreases.
- **Meanders** - bends of river; erosion on **outer (concave) bank**, deposition on **inner (convex) bank** = **point bar**.
- **Oxbow lake** - a meander neck is cut off in flood, leaving a crescent-shaped lake.
- **Flood plain** - broad flat area built by flood deposits on either side of the channel

(fertile).

- **Natural levees** - raised banks of coarse sediment deposited at channel edge during floods.
- **Braided channel** - river splits into many interwoven channels around bars (heavy load, variable flow).
- **Delta** - triangular (Greek letter Δ) deposit at river mouth: Krishna-Godavari delta, Ganga-Brahmaputra delta (largest). **Types:** arcuate (Nile), bird-foot (Mississippi), estuarine.
- **River terraces** - step-like flat surfaces left when the river cuts down into its old flood plain (rejuvenation).

4.5 Stages (Cycle) of a River - W. M. Davis

Feature	Youth	Maturity	Old age
Valley	V-shaped, narrow	Wider, open	Very broad flood plain
Gradient	Steep	Moderate	Almost flat
Work	Vertical erosion	Erosion + deposition	Deposition mostly
Features	Falls, rapids, gorge	Meanders, tributaries	Oxbow, levees, delta

- **Rejuvenation** - renewed down-cutting due to uplift of land or fall in sea level -> terraces, incised meanders, knick-points.

4.6 Engineering Importance of River Work

- **Dam & reservoir sites** - narrow gorges in strong rock in the upper course (e.g., Nagarjuna Sagar on River Krishna). Wide flood plains are poor sites for dams.
- **Bridges** - foundation must resist **scour** (erosion around piers); choose straight, stable reaches; avoid meander bends.
- **River training works** - guide banks, spurs, embankments, bank protection by riprap to stop lateral erosion.
- **Flood control** - study of flood plains, discharge records and levees.
- **Siltation** - sediment load fills reservoirs and reduces life; check catchment erosion.
- **Construction materials** - river sand, gravel and boulders for concrete; **alluvial soils** are poor foundations (soft, compressible, high water-table).
- **Water supply** - alluvium is the best aquifer; infiltration wells in river beds.
- **Pipelines, canals and roads** - route along stable terraces and ridges; avoid active

channel shifting.

Exam points (5 marks)

- 1. Define weathering and list types. 2. Explain river erosion, transportation and deposition. 3. Name 5 landforms of rivers with sketches. 4. Write two case studies showing importance of geology.

4.7 Important Terms - Quick Revision

- **Weathering** = in-situ breakdown; **Erosion** = wearing away with removal; **Denudation** = both together; **Gradation** = levelling.
- **Regolith** - loose cover of weathered rock; **Soil** - regolith with organic matter that supports plants.
- **Residual soil** stays at place; **transported soil** (alluvial, aeolian, glacial, marine, colluvial) is carried by agents.
- **Hydraulic action, abrasion, attrition, solution** = the four erosion mechanisms of a river.
- **Meander** - erosion outside, deposition inside; **delta** - deposition at mouth; **levee** - deposition at edges.

5. Short Answer Revision (important questions)

1. **Define engineering geology.** Application of geological principles to investigation, design, construction and maintenance of civil engineering works.
2. **Why is weathering studied by civil engineers?** It reduces strength and increases permeability of rock; decides depth of foundation and quality of construction material.
3. **Difference between weathering and erosion?** Weathering is in-situ breakdown; erosion includes removal by moving agents.
4. **Name the three types of weathering.** Physical, chemical and biological.
5. **What is exfoliation?** Peeling of curved shells from rock surface due to temperature change / unloading.
6. **Which process converts feldspar into clay?** Hydrolysis.
7. **Name any four branches of geology.** Mineralogy, petrology, structural geology,

hydrogeology, geomorphology.

8. **What is a delta?** Fan or triangle-shaped alluvial deposit at river mouth when velocity falls.
9. **Define meander and oxbow lake.** Bend in a river; a cut-off abandoned meander loop.
10. **Why are dam sites chosen in the upper course?** Narrow valley, strong rock, large head and small dam length = low cost.

Last-minute tips

- - Always draw a neat labelled sketch for rivers (V-valley, meander, delta).
- - For case studies write: place - year - type of structure - geological cause - lesson.
- - Use words like in-situ, hydrolysis, abrasion, attrition, competence, aggradation.